16.1-Random_Forest_Classification

August 17, 2025

1 Random Forest Classification Implementation

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import plotly.express as px
import warnings
warnings.filterwarnings('ignore')
import seaborn as sns
```

[2]:
```
df = pd.read_csv('../0-Dataset/Travel.csv')
df.head()
```

[2]:

	CustomerID	ProdTaken	Age	TypeofContact	CityTier	DurationOfPitch
0	200000	1	41.0	Self Enquiry	3	6.0
1	200001	0	49.0	Company Invited	1	14.0
2	200002	1	37.0	Self Enquiry	1	8.0
3	200003	0	33.0	Company Invited	1	9.0
4	200004	0	NaN	Self Enquiry	1	8.0

	Occupation	Gender	NumberOfPersonVisiting	NumberOfFollowups
0	Salaried	Female	3	3.0
1	Salaried	Male	3	4.0
2	Free Lancer	Male	3	4.0
3	Salaried	Female	2	3.0
4	Small Business	Male	2	3.0

	ProductPitched	PreferredPropertyStar	MaritalStatus	NumberOfTrips
0	Deluxe	3.0	Single	1.0
1	Deluxe	4.0	Divorced	2.0
2	Basic	3.0	Single	7.0
3	Basic	3.0	Divorced	2.0
4	Basic	4.0	Divorced	1.0

	Passport	PitchSatisfactionScore	OwnCar	NumberOfChildrenVisiting
0	1	2	1	0.0

```
1          0                       3        1                       2.0
2          1                       3        0                       0.0
3          1                       5        1                       1.0
4          0                       5        1                       0.0

  Designation  MonthlyIncome
0     Manager        20993.0
1     Manager        20130.0
2   Executive        17090.0
3   Executive        17909.0
4   Executive        18468.0
```

1.1 Data Cleaning

[3]:
```
df.isnull().sum()
```

[3]:
```
CustomerID                  0
ProdTaken                   0
Age                       226
TypeofContact              25
CityTier                    0
DurationOfPitch           251
Occupation                  0
Gender                      0
NumberOfPersonVisiting      0
NumberOfFollowups          45
ProductPitched              0
PreferredPropertyStar      26
MaritalStatus               0
NumberOfTrips             140
Passport                    0
PitchSatisfactionScore      0
OwnCar                      0
NumberOfChildrenVisiting   66
Designation                 0
MonthlyIncome             233
dtype: int64
```

[4]:
```
df['Gender'].value_counts()
```

[4]:
```
Gender
Male       2916
Female     1817
Fe Male     155
Name: count, dtype: int64
```

[5]:
```
df['Gender'] = df['Gender'].replace('Fe Male', 'Female')
```

[6]:
```
df['Gender'].value_counts()
```

[6]:
```
Gender
Male      2916
Female    1972
Name: count, dtype: int64
```

[7]:
```
df['MaritalStatus'].value_counts()
```

[7]:
```
MaritalStatus
Married      2340
Divorced      950
Single        916
Unmarried     682
Name: count, dtype: int64
```

[8]:
```
df['MaritalStatus'] = df['MaritalStatus'].replace('Single', 'Unmarried')
```

[9]:
```
df['MaritalStatus'].value_counts()
```

[9]:
```
MaritalStatus
Married      2340
Unmarried    1598
Divorced      950
Name: count, dtype: int64
```

[10]:
```
df['TypeofContact'].value_counts()
```

[10]:
```
TypeofContact
Self Enquiry       3444
Company Invited    1419
Name: count, dtype: int64
```

[11]:
```
df.head()
```

[11]:

	CustomerID	ProdTaken	Age	TypeofContact	CityTier	DurationOfPitch	\
0	200000	1	41.0	Self Enquiry	3	6.0	
1	200001	0	49.0	Company Invited	1	14.0	
2	200002	1	37.0	Self Enquiry	1	8.0	
3	200003	0	33.0	Company Invited	1	9.0	
4	200004	0	NaN	Self Enquiry	1	8.0	

	Occupation	Gender	NumberOfPersonVisiting	NumberOfFollowups	\
0	Salaried	Female	3	3.0	
1	Salaried	Male	3	4.0	
2	Free Lancer	Male	3	4.0	
3	Salaried	Female	2	3.0	
4	Small Business	Male	2	3.0	

```
  ProductPitched  PreferredPropertyStar MaritalStatus  NumberOfTrips  \
0         Deluxe                    3.0     Unmarried            1.0
1         Deluxe                    4.0      Divorced            2.0
2          Basic                    3.0     Unmarried            7.0
3          Basic                    3.0      Divorced            2.0
4          Basic                    4.0      Divorced            1.0

   Passport  PitchSatisfactionScore  OwnCar  NumberOfChildrenVisiting  \
0         1                       2       1                       0.0
1         0                       3       1                       2.0
2         1                       3       0                       0.0
3         1                       5       1                       1.0
4         0                       5       1                       0.0

  Designation  MonthlyIncome
0     Manager        20993.0
1     Manager        20130.0
2   Executive        17090.0
3   Executive        17909.0
4   Executive        18468.0
```

[12]:
```
## Handling the nan values
features_with_na = [features for features in df.columns if df[features].
 ↪isnull().sum()>=1]
for feature in features_with_na:
        print(feature, np.round(df[feature].isnull().mean()*100, 5), '% missing␣
 ↪values')
```

```
Age 4.62357 % missing values
TypeofContact 0.51146 % missing values
DurationOfPitch 5.13502 % missing values
NumberOfFollowups 0.92062 % missing values
PreferredPropertyStar 0.53191 % missing values
NumberOfTrips 2.86416 % missing values
NumberOfChildrenVisiting 1.35025 % missing values
MonthlyIncome 4.76678 % missing values
```

[13]:
```
df[features_with_na].select_dtypes(exclude='object').describe()
```

[13]:

	Age	DurationOfPitch	NumberOfFollowups	PreferredPropertyStar	\
count	4662.000000	4637.000000	4843.000000	4862.000000	
mean	37.622265	15.490835	3.708445	3.581037	
std	9.316387	8.519643	1.002509	0.798009	
min	18.000000	5.000000	1.000000	3.000000	
25%	31.000000	9.000000	3.000000	3.000000	
50%	36.000000	13.000000	4.000000	3.000000	
75%	44.000000	20.000000	4.000000	4.000000	

```
max      61.000000       127.000000           6.000000                5.000000

       NumberOfTrips  NumberOfChildrenVisiting  MonthlyIncome
count    4748.000000               4822.000000    4655.000000
mean        3.236521                  1.187267   23619.853491
std         1.849019                  0.857861    5380.698361
min         1.000000                  0.000000    1000.000000
25%         2.000000                  1.000000   20346.000000
50%         3.000000                  1.000000   22347.000000
75%         4.000000                  2.000000   25571.000000
max        22.000000                  3.000000   98678.000000
```

[14]:
```
df.Age.fillna(df.Age.median(), inplace =True)
df.TypeofContact.fillna(df.TypeofContact.mode()[0], inplace=True)
df.DurationOfPitch.fillna(df.DurationOfPitch.median(), inplace=True)
df.NumberOfFollowups.fillna(df.NumberOfFollowups.mode()[0], inplace=True)
df.PreferredPropertyStar.fillna(df.PreferredPropertyStar.mode()[0],␣
 ↪inplace=True)
df.NumberOfTrips.fillna(df.NumberOfTrips.median(), inplace=True)
df.NumberOfChildrenVisiting.fillna(df.NumberOfChildrenVisiting.mode()[0],␣
 ↪inplace=True)
df.MonthlyIncome.fillna(df.MonthlyIncome.median(), inplace=True)
```

[15]:
```
df.isnull().sum()
```

[15]:
```
CustomerID                0
ProdTaken                 0
Age                       0
TypeofContact             0
CityTier                  0
DurationOfPitch           0
Occupation                0
Gender                    0
NumberOfPersonVisiting    0
NumberOfFollowups         0
ProductPitched            0
PreferredPropertyStar     0
MaritalStatus             0
NumberOfTrips             0
Passport                  0
PitchSatisfactionScore    0
OwnCar                    0
NumberOfChildrenVisiting  0
Designation               0
MonthlyIncome             0
dtype: int64
```

[16]:
```
df.drop('CustomerID', axis=1, inplace=True)
```

1.2 Feature Engineering

[17]:
```
df.head()
```

[17]:

	ProdTaken	Age	TypeofContact	CityTier	DurationOfPitch
0	1	41.0	Self Enquiry	3	6.0
1	0	49.0	Company Invited	1	14.0
2	1	37.0	Self Enquiry	1	8.0
3	0	33.0	Company Invited	1	9.0
4	0	36.0	Self Enquiry	1	8.0

	Occupation	Gender	NumberOfPersonVisiting	NumberOfFollowups
0	Salaried	Female	3	3.0
1	Salaried	Male	3	4.0
2	Free Lancer	Male	3	4.0
3	Salaried	Female	2	3.0
4	Small Business	Male	2	3.0

	ProductPitched	PreferredPropertyStar	MaritalStatus	NumberOfTrips
0	Deluxe	3.0	Unmarried	1.0
1	Deluxe	4.0	Divorced	2.0
2	Basic	3.0	Unmarried	7.0
3	Basic	3.0	Divorced	2.0
4	Basic	4.0	Divorced	1.0

	Passport	PitchSatisfactionScore	OwnCar	NumberOfChildrenVisiting
0	1	2	1	0.0
1	0	3	1	2.0
2	1	3	0	0.0
3	1	5	1	1.0
4	0	5	1	0.0

	Designation	MonthlyIncome
0	Manager	20993.0
1	Manager	20130.0
2	Executive	17090.0
3	Executive	17909.0
4	Executive	18468.0

[18]:
```
df['TotalVisiting'] =␣
↪df['NumberOfChildrenVisiting']+df['NumberOfPersonVisiting']
df.drop(['NumberOfChildrenVisiting', 'NumberOfPersonVisiting'], axis=1,␣
↪inplace=True)
```

[19]:
```
# get all the numeric feature
num_features= [feature for feature in df.columns if df[feature].dtype !='O']
print('Number of Numerical Features: ', len(num_features))
```

```
Number of Numerical Features:  12
```

[20]:
```
# get all the categorical feature
num_features= [feature for feature in df.columns if df[feature].dtype =='O']
print('Number of Categorical Features: ', len(num_features))
```

```
Number of Categorical Features:  6
```

[21]:
```
# get all the discrete feature
discrete_features= [feature for feature in num_features if len(df[feature].
 ↪unique()) <=25]
print('Number of Discrete Features: ', len(discrete_features))
```

```
Number of Discrete Features:  6
```

[22]:
```
# Continuous Features
continuous_features = [feature for feature in num_features if feature not in␣
 ↪discrete_features]
print('number of Continuous Features: ', len(continuous_features))
```

```
number of Continuous Features:  0
```

2 Model Building

[23]:
```
from sklearn.model_selection import train_test_split
X = df.drop(['ProdTaken'], axis=1)
y = df['ProdTaken']
```

[24]:
```
y.value_counts()
```

[24]:
```
ProdTaken
0    3968
1     920
Name: count, dtype: int64
```

[25]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
 ↪random_state=42)
X_train.shape, X_test.shape
```

[25]:
```
((3910, 17), (978, 17))
```

[26]:
```
# Create Column Transformer with 3 types of transformers

cat_features = X.select_dtypes(include='object').columns
num_features = X.select_dtypes(exclude='object').columns
```

```
# we use the column Transformer for transform the columns with combining␣
 ↪multiple techniques

from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer

numeric_transformer = StandardScaler()
oh_transformer = OneHotEncoder(drop = 'first') # only (n-1) columns are required

preprocessor = ColumnTransformer(
        [
        ("OneHotEncoder", oh_transformer, cat_features),
        ('StandardScaler', numeric_transformer, num_features)
        ]
)
```

[27]:
```
print(preprocessor)
```

```
ColumnTransformer(transformers=[('OneHotEncoder', OneHotEncoder(drop='first'),
                                 Index(['TypeofContact', 'Occupation', 'Gender',
'ProductPitched',
       'MaritalStatus', 'Designation'],
      dtype='object')),
                                ('StandardScaler', StandardScaler(),
                                 Index(['Age', 'CityTier', 'DurationOfPitch',
'NumberOfFollowups',
       'PreferredPropertyStar', 'NumberOfTrips', 'Passport',
       'PitchSatisfactionScore', 'OwnCar', 'MonthlyIncome', 'TotalVisiting'],
      dtype='object'))])
```

[28]:
```
# Applying the Transformation
X_train = preprocessor.fit_transform(X_train)
X_test = preprocessor.transform(X_test)
```

[29]:
```
pd.DataFrame(X_train).head()
```

[29]:

	0	1	2	3	4	5	6	7	8	9	…	16	17
0	1.0	0.0	0.0	1.0	1.0	0.0	0.0	0.0	0.0	0.0	…	-0.7214	-1.020350
1	1.0	0.0	1.0	0.0	1.0	0.0	0.0	0.0	0.0	1.0	…	-0.7214	0.690023
2	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	…	-0.7214	-1.020350
3	1.0	0.0	1.0	0.0	1.0	1.0	0.0	0.0	0.0	1.0	…	-0.7214	-1.020350
4	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	…	-0.7214	2.400396

	18	19	20	21	22	23	24
0	1.284279	-0.725271	-0.127737	-0.632399	0.679690	0.782966	-0.382245
1	0.282777	-0.725271	1.511598	-0.632399	0.679690	0.782966	-0.459799
2	0.282777	1.771041	0.418708	-0.632399	0.679690	0.782966	-0.245196

```
3  1.284279 -0.725271 -0.127737 -0.632399  1.408395 -1.277194  0.213475
4 -1.720227 -0.725271  1.511598 -0.632399 -0.049015 -1.277194 -0.024889

         25
0 -0.774151
1  0.643615
2 -0.065268
3 -0.065268
4  2.061382

[5 rows x 26 columns]
```

3 Model Training

[30]:
```
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report,␣
↪confusion_matrix, precision_score, recall_score, f1_score, roc_auc_score,␣
↪roc_curve
```

3.1 An Efficient Way to Train Multiple Models

[31]:
```
# Defining all the models
models = {
        "Logistic Regression":LogisticRegression(),
        "Decision Tree":DecisionTreeClassifier(),
        "Random Forest":RandomForestClassifier(),
        "Gradient Boost":GradientBoostingClassifier()
}

# Training all the models
for i in range(len(list(models))):
        model = list(models.values())[i]
        model.fit(X_train, y_train)

        # Make Predictions
        y_train_pred = model.predict(X_train)
        y_test_pred = model.predict(X_test)

        # Training set Performance
        model_train_accuracy = accuracy_score(y_train, y_train_pred)
        model_train_confusion_matrix = confusion_matrix(y_train, y_train_pred)
        model_train_recall = recall_score(y_train, y_train_pred)
        model_train_f1 = f1_score(y_train, y_train_pred)
        model_train_precision = precision_score(y_train, y_train_pred)
```

```
        model_train_rocauc_score = roc_auc_score(y_train, y_train_pred)

        # Testing set Performance
        model_test_accuracy = accuracy_score(y_test, y_test_pred)
        model_test_confusion_matrix = confusion_matrix(y_test, y_test_pred)
        model_test_recall= recall_score(y_test, y_test_pred)
        model_test_f1 = f1_score(y_test, y_test_pred)
        model_test_precision= precision_score(y_test, y_test_pred)
        model_test_rocauc_score = roc_auc_score(y_test, y_test_pred)

        print(list(models.keys())[i])

        # showing the performances

        print('Model performance for Training Set')
        print("-Accuracy: {:.4f}".format(model_train_accuracy))
        print('- F1 score: {:.4f}'.format(model_train_f1))

        print('- Precision: {:.4f}'.format(model_train_precision))
        print('- Recall: {:.4f}'.format(model_train_recall))
        print('- Roc Auc Score: {:.4f}'.format(model_train_rocauc_score))
        print(f"-Confusion Matrix:\n {model_test_confusion_matrix}")



        print('----------------------------------')

        print('Model performance for Test set')
        print('- Accuracy: {:.4f}'.format(model_test_accuracy))
        print('- F1 score: {:.4f}'.format(model_test_f1))
        print('- Precision: {:.4f}'.format(model_test_precision))
        print('- Recall: {:.4f}'.format(model_test_recall))
        print('- Roc Auc Score: {:.4f}'.format(model_test_rocauc_score))
        print(f"-Confusion Matrix:\n {model_train_confusion_matrix}")


        print('='*35)
        print('\n')
```

```
Logistic Regression
Model performance for Training Set
-Accuracy: 0.8460
- F1 score: 0.4234
- Precision: 0.7016
- Recall: 0.3032
- Roc Auc Score: 0.6368
-Confusion Matrix:
 [[762  25]
```

```
 [135  56]]
---------------------------------
Model performance for Test set
- Accuracy: 0.8364
- F1 score: 0.4118
- Precision: 0.6914
- Recall: 0.2932
- Roc Auc Score: 0.6307
-Confusion Matrix:
 [[3087   94]
 [ 508  221]]
===================================


Decision Tree
Model performance for Training Set
-Accuracy: 1.0000
- F1 score: 1.0000
- Precision: 1.0000
- Recall: 1.0000
- Roc Auc Score: 1.0000
-Confusion Matrix:
 [[754  33]
 [ 50 141]]
---------------------------------
Model performance for Test set
- Accuracy: 0.9151
- F1 score: 0.7726
- Precision: 0.8103
- Recall: 0.7382
- Roc Auc Score: 0.8481
-Confusion Matrix:
 [[3181    0]
 [   0  729]]
===================================


Random Forest
Model performance for Training Set
-Accuracy: 1.0000
- F1 score: 1.0000
- Precision: 1.0000
- Recall: 1.0000
- Roc Auc Score: 1.0000
-Confusion Matrix:
 [[782   5]
 [ 71 120]]
---------------------------------
```

```
Model performance for Test set
- Accuracy: 0.9223
- F1 score: 0.7595
- Precision: 0.9600
- Recall: 0.6283
- Roc Auc Score: 0.8110
-Confusion Matrix:
 [[3181    0]
 [   0  729]]
==================================


Gradient Boost
Model performance for Training Set
-Accuracy: 0.8939
- F1 score: 0.6382
- Precision: 0.8756
- Recall: 0.5021
- Roc Auc Score: 0.7429
-Confusion Matrix:
 [[765  22]
 [116  75]]
----------------------------------
Model performance for Test set
- Accuracy: 0.8589
- F1 score: 0.5208
- Precision: 0.7732
- Recall: 0.3927
- Roc Auc Score: 0.6824
-Confusion Matrix:
 [[3129   52]
 [ 363  366]]
==================================
```

4 Hyperparameter Tuning

[32]:
```
rf_params = {
        "max_depth":[5,8,15,None, 10],
        "max_features":[5, 7, "auto", 8],
        "min_samples_split" : [2,8, 15, 20],
        "n_estimators":[100, 200, 500, 1000]
}
```

[33]:
```
rf_params
```

[33]:
```
{'max_depth': [5, 8, 15, None, 10],
 'max_features': [5, 7, 'auto', 8],
 'min_samples_split': [2, 8, 15, 20],
 'n_estimators': [100, 200, 500, 1000]}
```

[34]:
```
# Model list for hyperparameter tuning

randomcv_model = [
        ("RF", RandomForestClassifier(), rf_params)
]
```

[35]:
```
from sklearn.model_selection import RandomizedSearchCV

model_param = {}
for name, model, params in randomcv_model:
    random = RandomizedSearchCV(estimator=model,
                                param_distributions=params,
                                n_iter=100,
                                cv=3,
                                verbose=2,
                                n_jobs=-1)
    random.fit(X_train, y_train)
    model_param[name] = random.best_params_

for model_name in model_param:
    print(f"---------------- Best Params for {model_name} -------------------")
    print(model_param[model_name])
```

```
Fitting 3 folds for each of 100 candidates, totalling 300 fits

---------------- Best Params for RF -------------------
{'n_estimators': 500, 'min_samples_split': 2, 'max_features': 7, 'max_depth':
None}
```

[36]:
```
models={

    "Random Forest":RandomForestClassifier(n_estimators=500,min_samples_split=2,
                                           max_features=8,max_depth=15)
}

# Training all the models
for i in range(len(list(models))):
        model = list(models.values())[i]
        model.fit(X_train, y_train)

        # Make Predictions
        y_train_pred = model.predict(X_train)
        y_test_pred = model.predict(X_test)
```

```
        # Training set Performance
        model_train_accuracy = accuracy_score(y_train, y_train_pred)
        model_train_confusion_matrix = confusion_matrix(y_train, y_train_pred)
        model_train_recall = recall_score(y_train, y_train_pred)
        model_train_f1 = f1_score(y_train, y_train_pred)
        model_train_precision = precision_score(y_train, y_train_pred)
        model_train_rocauc_score = roc_auc_score(y_train, y_train_pred)

        # Testing set Performance
        model_test_accuracy = accuracy_score(y_test, y_test_pred)
        model_test_confusion_matrix = confusion_matrix(y_test, y_test_pred)
        model_test_recall= recall_score(y_test, y_test_pred)
        model_test_f1 = f1_score(y_test, y_test_pred)
        model_test_precision= precision_score(y_test, y_test_pred)
        model_test_rocauc_score = roc_auc_score(y_test, y_test_pred)

        print(list(models.keys())[i])

        # showing the performances

        print('Model performance for Training Set')
        print("-Accuracy: {:.4f}".format(model_train_accuracy))
        print('- F1 score: {:.4f}'.format(model_train_f1))

        print('- Precision: {:.4f}'.format(model_train_precision))
        print('- Recall: {:.4f}'.format(model_train_recall))
        print('- Roc Auc Score: {:.4f}'.format(model_train_rocauc_score))
        print(f"-Confusion Matrix:\n {model_test_confusion_matrix}")



        print('----------------------------------')

        print('Model performance for Test set')
        print('- Accuracy: {:.4f}'.format(model_test_accuracy))
        print('- F1 score: {:.4f}'.format(model_test_f1))
        print('- Precision: {:.4f}'.format(model_test_precision))
        print('- Recall: {:.4f}'.format(model_test_recall))
        print('- Roc Auc Score: {:.4f}'.format(model_test_rocauc_score))
        print(f"-Confusion Matrix:\n {model_train_confusion_matrix}")


        print('='*35)
        print('\n')
```

```
Random Forest
Model performance for Training Set
```

```
-Accuracy: 0.9995
- F1 score: 0.9986
- Precision: 1.0000
- Recall: 0.9973
- Roc Auc Score: 0.9986
-Confusion Matrix:
 [[782   5]
 [ 61 130]]
----------------------------------
Model performance for Test set
- Accuracy: 0.9325
- F1 score: 0.7975
- Precision: 0.9630
- Recall: 0.6806
- Roc Auc Score: 0.8371
-Confusion Matrix:
 [[3181    0]
 [   2  727]]
==================================
```

[37]:
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## Plot ROC AUC Curve
from sklearn.metrics import roc_auc_score,roc_curve
plt.figure()

# Add the models to the list that you want to view on the ROC plot
auc_models = [
{
    'label': 'Random Forest Classifier',
    'model': RandomForestClassifier(n_estimators=500,min_samples_split=2,
                                        max_features=8,max_depth=15),
    'auc':  0.8352
},

]
# create loop through all model
for algo in auc_models:
    model = algo['model'] # select the model
    model.fit(X_train, y_train) # train the model
# Compute False postive rate, and True positive rate
    fpr, tpr, thresholds = roc_curve(y_test, model.predict_proba(X_test)[:,1])
# Calculate Area under the curve to display on the plot
    plt.plot(fpr, tpr, label='%s ROC (area = %0.2f)' % (algo['label'],␣
  ↪algo['auc']))
# Custom settings for the plot
plt.plot([0, 1], [0, 1],'r--')
```

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plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('1-Specificity(False Positive Rate)')
plt.ylabel('Sensitivity(True Positive Rate)')
plt.title('Receiver Operating Characteristic')
plt.legend(loc="lower right")
plt.savefig("auc.png")
plt.show()
```

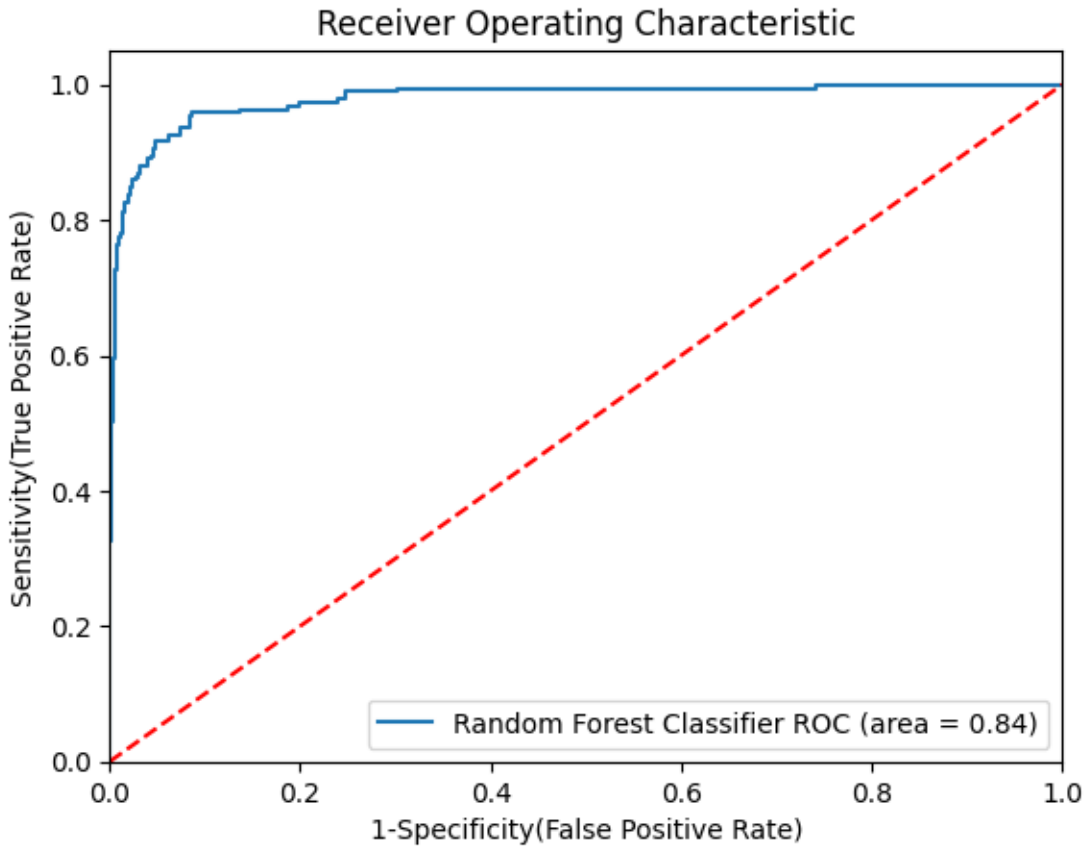

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